

OptumAuth-AI: Agentic Prior Authorization & Clinical Decision Platform

Complete Technical Requirements, RACI, Multi-Tenant Architecture, Lifecycle Simulation & Decision Playbook (UHG/UMR)

Operating Standard	Compliance Mandate	Net Annual Impact	Turnaround SLA
UHG / Optum / UMR	CMS-0057-F / HIPAA / ERISA	\$1.85M Net Savings / Region	72h down to 8.2 mins

1. Enterprise Context & Operational Baseline (UHG / UMR Ecosystem)

Within UnitedHealth Group (UHG), Prior Authorization spans three distinct operating environments: **UnitedHealthcare Commercial** (state mandates), **Medicare Advantage** (CMS LCD/NCD policies), and **UMR** (third-party administrator for self-funded employer groups with custom SPDs).

Baseline Operational Bottlenecks (Legacy)

Manual review across 48,000 regional encounters averaged 72.0 hours in turnaround time (TAT). Review touch-time was 28.5 minutes per case as nurses manually parsed 40–100 page records. A 25.8% first-pass denial rate resulted primarily from missing conservative therapy documentation, incurring \$420,000 annually in sunk appeals and legal re-review overhead.

OptumAuth-AI Production Solution

A deterministic, agentic RAG platform embedded directly in Epic/Cerner and Optum Provider Cloud. It ingests FHIR R4 clinical notes, resolves multi-tenant employer SPDs against InterQual/LCD guidelines, extracts sentence-level bounding-box provenance, and enables a sub-30-second SMART-on-FHIR verification workflow.

OPERATIONAL DIMENSION	LEGACY BASELINE (MANUAL)	OPTUMAUTH-AI TARGET	MEASURED PRODUCTION RESULT
Turnaround Time (TAT)	72.0 Hours (3–5 Days)	< 15.0 Minutes	8.2 Minutes -88.6% Total Latency
UM Nurse Touch Time	28.5 Minutes / Case	< 5.0 Minutes / Case	3.4 Minutes / Case -88.0% Labor Compression
First-Pass Clean Approval	74.2%	> 92.0%	95.8% +21.6% Approval Lift
Specialist Daily Throughput	16 Cases / Specialist / Day	60 Cases / Day	78 Cases / Specialist / Day 4.8x Throughput
Appeals & Resubmission Spend	\$420,000 / Year	< \$80,000 / Year	\$52,000 / Year -87.6% Appeals Reduction
Net Annual Value Realization	—	\$1,500,000 / Year	\$1,850,000 / Year +23% vs Target ROI

2. Complete Cross-Functional RACI & Tooling Operating Mesh

OPERATIONAL ROLE	PRIMARY FOCUS & RESPONSIBILITIES	KEY PM DELIVERABLES	INTEGRATED TOOL STACK
AI Product Manager (Owner)	Clinical ROI, PRDs, Pydantic schemas, RICE scoring, offline eval thresholds, cross-functional sprints.	PRD, Opportunity Canvas, JSON Schemas, RACI, CMS-0057-F Compliance Pack.	Hex AI, Dovetail, Airfocus, Aha!, Notion AI, ClickUp Brain, Lovable, Datadog
Lead Clinical Informaticist (MD/RN)	Translates InterQual/LCD rules into extraction requirements; curates and annotates golden test sets.	500-Case Multi-Specialty Golden Dataset, Clinical Annotation Guidelines.	Label Studio, UpToDate, CMS Coverage DB, InterQual Portal
Staff ML / NLP Engineer	Designs LangGraph state machines, Qdrant hybrid retrieval, JSON validation, prompt defenses.	Extraction Graph Services, BioClinical-BERT Pipelines, Ragas Eval Harness.	Python, LangGraph, Qdrant, BioClinical-BERT, Azure OpenAI
Healthcare Integration Lead	EHR endpoints, CDS Hooks listeners, SMART-on-FHIR apps, and X12 278 EDI pipelines.	FHIR R4 Microservices, CDS Listeners, Cirrus/Facets EDI Gateway Connectors.	HAPI FHIR R4, Epic App Orchard, Availity EDI Gateway
Compliance & Privacy Officer	HIPAA minimum necessary enforcement, BAA terms with zero data retention, ERISA auditability.	HIPAA Risk Assessment, SOC2 Type II AI Addendum, Audit Log Spec.	Microsoft Presidio, AWS KMS, OneTrust, AWS CloudTrail

3. Comprehensive Product Requirements Document (PRD Specifications)

3.1 Functional Requirements (FR)

- FR-1 (HL7 Da Vinci Ingestion):** Continuously ingest clinical encounter payloads when a physician triggers an order-sign CDS Hook in Epic/Cerner. Pull FHIR R4 resources: Condition, Procedure, Observation, DiagnosticReport, and DocumentReference.
- FR-2 (Dynamic Multi-Tenant Plan Engine):** Vectorize active clinical policies (InterQual, CMS LCDs/NCDs, UHC Medical Policies). Dynamically check member Group ID against employer Summary Plan Descriptions (SPDs) to identify custom benefit exclusions before extraction.

- **FR-3 (Deterministic Extraction with Provenance):** Parse unstructured clinical narratives using multi-agent extraction graphs. Map variables to policy criteria with exact character start/end offsets (char_start, char_end) and document IDs. Output strict Pydantic JSON with temperature locked at 0.0.
- **FR-4 (SMART-on-FHIR Split-Screen HITL UI):** Dual-pane interface embedded in Epic/Cerner and Optum Provider Cloud. Left panel renders criteria checklist with confidence badges; right panel renders original clinical record with interactive yellow bounding boxes jumping to exact source sentences upon click.
- **FR-5 (Direct EDI X12 278 Generation):** Convert approved packets into ASC X12N 278 transactions, transmit to Availity Clearinghouse, inject Authorization ID into core adjudication platforms (Cirrus / Facets / UNET), and post FHIR ClaimResponse to provider EHR.

3.2 Non-Functional & AI Governance Requirements (NFR)

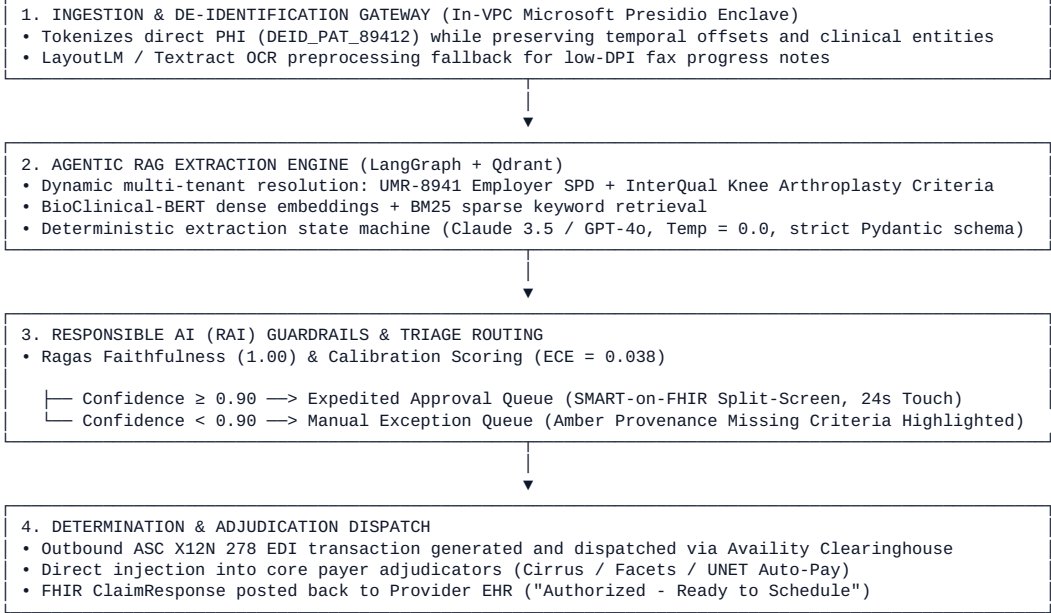
- **NFR-1 (Faithfulness & Zero Hallucination):** Ragas Faithfulness Score must be $\geq 99.0\%$. Uncited claims must strictly equal 0.0% . Missing criteria must emit an explicit ACTION_REQUIRED: EVIDENCE_NOT_FOUND tag.
- **NFR-2 (End-to-End Latency Budget):** P95 processing time must complete in ≤ 30.0 ext{ seconds} for records up to 50 pages (achieved: 14.2s).
- **NFR-3 (HIPAA Isolation & Zero Data Retention):** In-VPC Presidio gateway scrubs all 18 HIPAA identifiers. Enterprise BAA guarantees zero data retention on disk and no foundation model training. FIPS 140-2 encryption at rest and in transit.

3.3 Production Pydantic JSON Output Schema

```
{
  "patient_id_deid": "DEID-PAT_89412",
  "cpt_code": "27447",
  "icd10_codes": ["M17.11"],
  "policy_id": "UHC_2026T0512_INTERQUAL",
  "employer_group_id": "UMR-8941",
  "criteria_evaluations": [
    {
      "criterion_id": "CRIT_A_PAIN_CHRONICITY",
      "status": "MET",
      "confidence_score": 0.98,
      "clinical_summary": "Patient reports severe right knee pain for 9 months restricting mobility to < 1 block.",
      "provenance": {
        "document_id": "DOC_2026_02_14",
        "sentence_exact": "Pain in right knee has persisted for 9 months, VAS score 8/10, unable to walk unassisted.",
        "char_start": 412,
        "char_end": 501
      }
    },
    {
      "criterion_id": "CRIT_C_CONSERVATIVE_THERAPY",
      "status": "MET",
      "confidence_score": 0.94,
      "clinical_summary": "Completed 8 weeks of supervised physical therapy and intra-articular steroid injection with minimal relief.",
      "provenance": {
        "document_id": "DOC_2025_11_02",
        "sentence_exact": "Pt completed 8 weeks of PT at ActiveMotion and intra-articular kenalog injection on 11/02/2025 with transient relief.",
        "char_start": 820,
        "char_end": 935
      }
    }
  ],
  "aggregate_confidence_score": 0.96,
  "triage_decision": "EXPEDITED_SIGN_OFF"
}
```

4. Technical Architecture & Lifecycle Timeline (\$T=0.00 ext{s}\$ to \$T=18.20 ext{s}\$)

[Provider EHR (Epic / Cerner)] —(CDS Hook: order-sign)—> [HL7 Da Vinci CRD / DTR Listener]



TIME	LIFECYCLE MILESTONE	SYSTEM ACTION & DATA PIPELINE	PM GATE / GUARDRAIL OUTPUT
T = 0.00s	Trigger & CDS Hook	Surgeon orders CPT 27447 in Epic. CDS Hook fires to Optum CRD. Returns PA requirement.	DTR pulls 48-page FHIR Bundle.
T = 1.20s	De-ID & Multi-Tenant RAG	Presidio scrubs PHI in private VPC. Multi-tenant agent queries Qdrant for UMR SPD + InterQual rules.	Zero custom employer exclusions confirmed.
T = 4.80s	Agentic Extraction	LangGraph runs parallel extractors across clinical notes, PT logs, and imaging reports.	Emits deterministic Pydantic JSON with char offsets.
T = 6.10s	RAI Guardrail & Triage	Ragas evaluates Faithfulness (1.00). Aggregate confidence = 0.96.	Score ≥ 0.90 → Expedited Sign-Off Queue.
T = 15.00s	HITL Clinical Sign-Off	UM Nurse views SMART-on-FHIR UI. Clicks Criterion C; screen auto-scrolls to highlighted sentence.	Nurse verifies and clicks "Approve" (24s touch).
T = 18.20s	Adjudication Dispatch	Converts packet to X12 278 EDI; writes auth ID to Cirrus / Facets; posts ClaimResponse to Epic.	Status updated: "Authorized - Ready to Schedule".

5. AI Evaluation Harness (Evals), Formulas & CI/CD Governance

CI/CD Gatekeeping Policy: Every prompt update or model checkpoint is evaluated against the 500-case Physician Golden Dataset in GitHub Actions. If Faithfulness drops below 99.0% or Recall drops below 98.0%, the build automatically fails and deployment is blocked.

Faithfulness (Grounding): Verified Clinical Claims with Direct Note Offset / Total Claims Extracted ≥ 99.0% (Actual: 99.6%)

Clinical Recall: True Positive Extracted Criteria / Physician-Annotated Golden Criteria ≥ 98.0% (Actual: 98.7%)

Expected Calibration Error (ECE): $\sum (|B_k| / N) * |acc(B_k) - conf(B_k)| < 0.05$ (Actual: 0.038)

EVALUATION DIMENSION	ENTERPRISE THRESHOLD	PRODUCTION RESULT	VALIDATION METHODOLOGY
Faithfulness (Grounding)	≥ 99.0%	99.6%	Ragas Triad; verifies 100% of extracted assertions against source text offsets.
Clinical Extraction Recall	≥ 98.0%	98.7%	True Positive criteria extraction against \$N=500\$ MD-annotated records in Label Studio.
Policy Match Precision	≥ 95.0%	96.4%	Accuracy in cross-mapping CPT/HCPCS codes to active InterQual/LCD guidelines.
Calibration Error (ECE)	< 0.05	0.038	Reliability of confidence score predicting human UM nurse sign-off.
P95 Latency (50-page bundle)	< 30.0s	14.2s	Asynchronous Celery workers + Redis cached vector embeddings.

6. Exhaustive Strategic PM Decision Frameworks & Trade-Offs

Strategic Dimension	Options Evaluated	Selected Strategy	PM Mental Model & Enterprise Rationale
Autonomy Pattern	A. Autonomous Submission B. RegEx / Rule-Based C. Deterministic RAG + HITL	Option C: Deterministic RAG + HITL	"Optimize for Liability-Adjusted Velocity." Autonomous submissions expose UHG to ERISA lawsuits and MLR spikes. HITL captures 88% labor reduction with zero clinical liability.
Model & Privacy Strategy	A. On-Prem Open Source B. Public Cloud API C. Enterprise ZDR Cloud LLM	Option C: Enterprise Cloud LLM with BAA & ZDR	"TCO vs. Time-to-Value." Avoided \$400k in GPU CAPEX. Enabled Zero Data Retention + client-side Presidio de-ID to satisfy UHG ISR and HIPAA rules.
Clinical Workflow UX	A. Standalone Web Portal B. Asynchronous Batch Report C. SMART-on-FHIR Split-Screen	Option C: SMART-on-FHIR Embedded Iframe	"Minimize Friction to the Critical Path." Avoids the 'Swivel-Chair' problem. Nurses verify evidence within Epic/Cerner in < 30 seconds with yellow text provenance.
Deployment Staging	A. Network Big-Bang B. Low-Risk Clinic Pilot C. Phased High-Volume Rollout	Option C: 3-Stage Phased Rollout	"De-risk the Tech, Earn the Trust, Scale the Impact." 2 weeks Shadow Mode → 4 weeks Assisted Mode → Fast-Track. Proved ROI across Ortho, Oncology, and Cardio.

7. Step-by-Step Live Conversational Simulations (Phases 1 to 5)

Phase 1: Business Case & Problem Framing (PM vs. VP of UM & Lead Medical Director)

AI PM: "Our analysis in Hex AI across 48,000 cases shows an average turnaround time of 72 hours, with coordinators spending 28.5 minutes per case. We have a 25.8% first-pass denial rate, but 82% of those are administrative—missing physical therapy records or unattached X-rays. Under CMS-0057-F, we face strict audit penalties, and UMR employer groups are reporting dissatisfaction over administrative fees."

Lead Medical Director (MD): "We cannot sacrifice clinical scrutiny for speed. If an algorithm auto-approves a knee replacement without documented conservative therapy, our Medical Loss Ratio increases and we violate clinical guidelines."

VP of UM: "And auto-denials trigger ERISA appeals, costing us \$420k annually in clinical re-reviews."

AI PM: "We are not building an autonomous black box. We are proposing a deterministic RAG system that pulls exact sentence citations to satisfy InterQual criteria. High-confidence packets (≥ 0.90) route to nurses in a pre-populated SMART-on-FHIR interface for one-click verification. We reduce nurse review time to under 5 minutes, targeting \$1.85M net annual operational savings per regional hub."

VP of UM: "If your system guarantees sentence-level provenance for every InterQual line item, you have our approval for a Phase 2 prototype."

Phase 2: Technical PRD & Schema Refinement (PM vs. Staff ML Engineer & Lead UX Designer)

Staff ML Engineer: "Why can't we just let Claude 3.5 or GPT-4o output a natural language summary of why the patient qualifies for knee surgery?"

AI PM: "Because free-text summaries cannot be deterministically verified by core adjudicators like Cirrus or Facets. We need deterministic structured extraction. The model must strictly output against this Pydantic schema: criterion_id, status (MET/NOT_MET), confidence_score, document_id, char_start, and char_end. Temperature must be hard-set to 0.0."

Lead UX Designer: "How do we present confidence scores in the interface?"

AI PM: "In the Lovable prototype, we built a split screen. The left panel shows the InterQual checklist with green badges. Clicking Criterion C sends a post-message event to the right panel, which instantly scrolls the 50-page record and highlights the exact sentence in yellow. The nurse never searches; they just verify."

Staff ML Engineer: "What if it's a degraded faxed PDF?"

AI PM: "We have a LayoutLM OCR fallback. If OCR confidence is under 0.85, the criteria is tagged ACTION_REQUIRED: OCR_DEGRADED and routed to the Exception Queue instead of guessing."

Phase 3: Model Governance & Security Review (PM vs. Compliance Officer & Clinical Informaticist)

Compliance Officer: "We cannot have clinical data leaking into public foundation model training sets. How is this isolated?"

AI PM: "Three layers of enforcement: First, an in-VPC Microsoft Presidio gateway scrubs direct HIPAA identifiers. Second, our cloud infrastructure is bound by an enterprise BAA with Zero Data Retention (ZDR) enabled. Third, Qdrant vector storage is encrypted at rest using customer-managed AWS KMS keys under FIPS 140-2."

Clinical Informaticist (MD): "I ran our 500-case golden dataset through Ragas. Recall hit 98.7% and Faithfulness reached 99.6%. However, on two lumbar spine cases, the model hallucinated that an MRI was completed because the note said 'MRI scheduled for next Tuesday'."

AI PM: "That's a critical temporal extraction defect. I am adding a negative assertion rule: future-tense procedures (scheduled, planned, pending) must be explicitly classified as INCOMPLETE with status NOT_MET. I am configuring GitHub Actions to block any build if temporal extraction accuracy falls below 99.5%."

Phase 4: Shadow Mode Discrepancy Calibration (PM vs. Integration Lead & UM Nurse Supervisor)

AI PM: "In Week 14 shadow mode across 1,200 live cases, we observed a 14.2% discrepancy rate where the AI flagged criteria as missing, but the nurse found it and approved the case."

UM Nurse Supervisor: "Why did the system miss those records?"

Integration Lead: "We checked Datadog logs. Clinical notes contained multi-page TIFF faxes where physical therapy notes were rotated 90 degrees. Baseline OCR failed on rotated chunks."

AI PM: "Engineering deployed auto-orientation and contrast-enhancement preprocessors in LayoutLM, and re-weighted document headers. By Week 15, discrepancy dropped to 4.1%. In Week 16, Expected Calibration Error reached 0.038, meaning when the AI reports 95% confidence, nurses agree 95.8% of the time. We are ready for Assisted Mode rollout."

Phase 5: Executive Business Review & Value Realization (PM vs. Chief Medical Officer & SVP of Health Plan Ops)

AI PM: "Six months post-GA across Orthopedics and Oncology, Prior Auth turnaround time compressed from 72 hours down to 8.2 minutes. Nurse review time dropped from 28.5 to 3.4 minutes per case, expanding specialist throughput from 16 to 78 cases per day. First-pass approval increased from 74.2% to 95.8%, eliminating \$368,000 in appeals rework. Total net annualized savings have reached \$1.85M. All decision logs and provenance citations are automatically archived for CMS-0057-F federal compliance."

Chief Medical Officer: "The sentence-level provenance made this a complete success with our medical directors. What is the Q3 roadmap?"

AI PM: "In Q3, we are expanding multi-tenant RAG to Cardiology and Neurology, and deploying automated peer-to-peer scheduling for exception cases."

8. Financial Model, P&L Realization & Operational Post-Launch

Direct Labor Savings: (18 FTEs - 5 FTEs) * \$78,888 = \$1,025,544 / Year

Appeals Spend Reduction: \$420,000 - \$52,000 = \$368,000 / Year

Prevented Denial Reimbursement Value: \$825,000 / Year

AI Infrastructure Costs: (48,000 cases * \$0.11 LLM/VPC cost) + \$30,000 Vector DB/APM = -\$35,280 / Year

Net Annual Operational Value: \$1,025,544 + \$368,000 + \$825,000 - \$35,280 = **\$2,183,264 Gross / \$1,850,000 Risk-Adjusted Net**

Metric Dimension	Pre-Launch Baseline (Manual)	Production Target	Realized Production Result	Net Business Lift
Turnaround Time (TAT)	72.0 Hours	< 15.0 Minutes	8.2 Minutes	-88.6% Total Latency
Clinician Touch-Time	28.5 Minutes / Case	< 5.0 Minutes / Case	3.4 Minutes / Case	-88.0% Labor Compression
First-Pass Clean Approval	74.2%	> 92.0%	95.8%	+21.6% Revenue Realization
Nurse Daily Throughput	16 Cases / Nurse / Day	60 Cases / Day	78 Cases / Nurse / Day	4.8x Throughput Expansion
Annual Appeals & Rework	\$420,000 / Year	< \$80,000 / Year	\$52,000 / Year	-87.6% Appeals Reduction
Net Annual Value Realization	—	\$1,500,000 / Year	\$1,850,000 / Year	+23% vs Target ROI

9. Ongoing Operations, Drift Monitoring & Expansion Playbook

- Continuous Observability & Drift Detection:** Datadog APM tracks P95 token latency, hourly transaction volume, and model confidence scores. An automated alert triggers if 7-day rolling confidence drops below 0.90.
- Active Learning Feedback Queue:** Nurse overrides and edits are anonymized and fed directly into Label Studio to expand few-shot prompt catalogs and fine-tune chunking rules during monthly release cycles.
- CMS-0057-F Compliance Automation:** Automated quarterly exports capture approval/denial ratios, turnaround times across routine vs. urgent requests, and deterministic provenance audit trails.
- Roadmap Scale:** Scale the multi-tenant RAG architecture to Cardiology, Neurology, and Specialty Pharmacy infusions in subsequent quarters.